

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence and Data Science) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIOE604A - BIG DATA ANALYTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Mapper implementations are passed the JobConf for the job via the _____ method. a) JobConfigure.configure b) JobConfigurable.configure c) JobConfigurable.configurable d) None of the mentioned
- 2 The Mapper implementation processes one line at a time via _____ method. a) map b) reduce c) mapper d) reduce
- 3 What was Hadoop written in? a) Java (software platform) b) Perl c) Java (programming language) d) Lua (programming language)
- 4 Above the file systems comes the _____ engine, which consists of one Job Tracker, to which client applications submit MapReduce jobs. a) MapReduce b) Googlec) Functional programming d) Facebook
- 5 _____ has the world's largest Hadoop cluster. a) Apple b) Datamatics c) Facebook d) None of the mentioned
- 6 What does the 'Velocity' characteristic of Big Data refer to? (a) The amount of data (b) The speed at which data is generated (c) The different types of data (d) The source of data
- 7 Which of the following is NOT a Big Data tool? (a) Hadoop (b) Qubole (c) Spark (d) SQL
- 8 What are the main components of Big Data? (a) MapReduce (b) HDFS (c) YARN (d) All of these
- 9 Which type of data refers to information that is generated in real-time and requires immediate processing? a) Structured data b) Semi-structured data c) Unstructured data d) Streaming data
- 10 _____ is general-purpose computing model and runtime system for distributed data analytics. (a) Mapreduce (b) Drill (c) Oozie (d) None of the above
- 11 The word 'Big data' was coined by (a) Roger Mougallas (b) John Philips (c) Simon Woods (d) Martin Green
- 12 What is the purpose of MapReduce in Hadoop? a) To create maps of geographical locations b) To visualize data on maps c) To process and analyze large datasets in parallel d) To generate reports from data

Q2. Solve the following.

- A) Define Big Data. Explain the Evolution of Big Data and their characteristics 6
- B) Illustrate the Hadoop core components with neat diagram 6

Q3. Solve the following.

- A) Describe the MapReduce execution steps with neat diagram 6
- B) Illustrate main features and Architecture of Hive with neat diagram 6

Q4. Solve Any Two of the following.

- A) What is Hadoop? Explain Hadoop eco-system with neat diagram 6
- B) Explain with neat diagram HDFS Components. 6
- C) Write short note on Apache hive 6

Q5. Solve Any Two of the following.

- A) What is HIVE? Explain HIVE Architecture 6
- B) What is PIG in Big Data? Explain the features of PIG 6
- C) Define key-value store with examples. What are the advantages of key-value store? 6

Q6. Solve Any Two of the following.

- A) Highlight Big data Analytics Applications with one case study 6
- B) Explain Apache kafka with concept and architecture 6
- C) write a short note on Kerberos authentication 6

*** End ***